

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-18 16:26 UTC

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Generated 2026-05-18 16:26 UTC

- Cycles recorded: **784**
- Time span: 594.8h (~1.32 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	744
BARRIER_HIT	20
FALSIFIED	16
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	40
Schur-Weyl Duality	28
Free Probability Theory	19
Representation Theory of Symmetric Groups	18
Matroid Theory	16
Free Probability	14
Additive Combinatorics	13
Noncommutative L^p Geometry	10
Random Matrix Theory	8
Polymatroid Theory	8
Algebraic Geometry	7
Spectral Graph Theory	7
Finite Geometry	6
Plethysm Theory	6
Persistent Homology	5
Noncommutative Harmonic Analysis	5
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier Analysis on Boolean Functions	4

Field	Cycles
Algebraic Geometry over Finite Fields	4
Fourier Analysis of Boolean Functions	4
Algebraic Geometry of Secant Varieties	4
FOURIER_ANALYSIS	4
Noncommutative Geometry	4
Fourier analysis of boolean functions	3
Algebraic Topology	3
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-18 06:47 UTC	INCONCLUSIVE	Bourgain Noise-Sensitivity Mass of Clause-Falsification Polynomialia
2026-05-18 07:28 UTC	INCONCLUSIVE	Generated Algebra Dimension of Layer Operators Bounds IP_2 Read-T
2026-05-18 08:08 UTC	INCONCLUSIVE	Szegedy Quantum-Walk Phase Gap Lower-Bounds Tseitin Tree-Resoluti
2026-05-18 08:28 UTC	INCONCLUSIVE	Halász L^2 Spectrum Discrepancy Lower-Bounds Sign-Matrix Rigidity
2026-05-18 08:56 UTC	INCONCLUSIVE	Mostar Index of Gate-Adjacency Graph Bounds $ACC^0[m]$ Size for MOD
2026-05-18 09:23 UTC	INCONCLUSIVE	Patience-Sort LIS of XOR-Lifted Row Permutations Bounds CC^D
2026-05-18 09:54 UTC	INCONCLUSIVE	Effective Resistance of Charged Vertex Pairs Bounds Tseitin DPLL
2026-05-18 10:26 UTC	INCONCLUSIVE	$p=3$ Path-Family Modulus Lower-Bounds Tseitin Tree-Resolution
2026-05-18 11:02 UTC	INCONCLUSIVE	Viennot Heap Radius of Clause-Conflict Graph Bounds Frege Depth
2026-05-18 11:29 UTC	INCONCLUSIVE	Lee-Yang Zero Cluster Angle of Cut Polynomial Bounds Max-Cut SoS-
2026-05-18 11:55 UTC	INCONCLUSIVE	Additive Energy of Out-Degree Sequence Bounds $ACC^0[2]$ MOD_3 Size

Time	Verdict	Title
2026-05-18 12:32 UTC	INCONCLUSIVE	SBM Detectability Gap of Literal Conflict Graph Bounds DPLL Time
2026-05-18 13:00 UTC	INCONCLUSIVE	Fiedler Participation Entropy Lower-Bounds Tseitin Tree-Resolutio
2026-05-18 13:47 UTC	INCONCLUSIVE	Hypercontractive Term-Pair Contraction Defect Bounds Monotone CLI
2026-05-18 16:26 UTC	INCONCLUSIVE	Lovász-Theta of Term-Conflict Graph Bounds Monotone k-CLIQUE DNF

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```